

P1 Endo/Repro Practice Questions

Week 1

Questions

1. Which of the following antibodies are indicative of Type 1 Diabetes mellitus (T1DM) ?
 - a. Glutamic Acid Decarboxylase (GAD)
 - b. Zinc transporter (ZnT8)
 - c. Anti-M antibodies
 - d. A and B
 - e. All of the above
2. Which of the following best describes pathophysiological changes in Type 2 Diabetes mellitus (T2DM) ?
 - a. GLP1 decreases while GIP increases
 - b. Both GLP1 and GIP decrease
 - c. Both GLP1 and GIP increase
 - d. None of the above
3. You recently diagnosed Mark, an 18-year-old patient with T1DM. He works shifts in a factory and frequently skips meals because of his work schedule. What would be the best MDI insulin regimen for him considering the nature of his work?
 - a. 3-injections per day (1 long-acting + 2 fast-acting)
 - b. 4 injections per day (3 fast-acting + 1 long-acting)
 - c. 4 injections per day (2 long-acting + 2 fast-acting)
 - d. None of the above
4. Which of the following is FALSE regarding the diagnosis of T1DM
 - a. Polyuria and polydipsia are diagnostic symptoms of T1DM
 - b. A random blood glucose of > 11.1 is sufficient for the diagnosis of T1DM in a symptomatic patient
 - c. You must do a fasting plasma glucose and/or HbA1c to confirm the diagnosis of T1DM in a symptomatic patient
 - d. None of the above
5. Which of the following is TRUE regarding T1DM and T2DM?
 - a. T1DM and T2DM are both caused by absolute insulin deficiency
 - b. T2DM often presents as an acute condition
 - c. Patients with T2DM are more likely to have other manifestations of metabolic syndrome
 - d. None of the above is true
6. Which of the following is TRUE regarding the risk of T2DM
 - a. A child born to a parent with T2DM has a higher risk of developing T2DM
 - b. If a twin develops T2DM, their monozygotic twin has a 90% risk of developing T2DM
 - c. The risk of developing T2DM is the same for monozygotic and non-monozygotic twins
 - d. A and B are both true
 - e. None of the above

7. A patient presents to the ER with polydipsia and polyuria that have been going on for several days. Blood glucose is 30 mmol. pH is 7.4. Creatinine is elevated and the patient is looking very sick. The patient's mother has a history of T2DM. What is the most likely diagnosis?
 - a. DKA
 - b. Septic shock
 - c. HHS
 - d. None of the above
8. A patient with T1DM presents to the ophthalmologist for retinal evaluation. Retinal exam reveals the presence of new blood vessels in the retina. What is the most appropriate next step?
 - a. The patient needs to be monitored closely as they have pre-proliferative retinopathy
 - b. No further action is needed
 - c. Refer the patient to hematology
 - d. The patient requires treatment for proliferative retinopathy and effective glycemic control.
9. Which of the following is TRUE regarding the findings of DCCT trial ?
 - a. Intensive BG control is associated with a reduction retinopathy only
 - b. Intensive BG control is associated with a reduction in retinopathy and preventing worsening of existing retinopathy
 - c. Intensive BG control is associated with a reduction retinopathy, neuropathy, and microalbuminuria
 - d. Both B and C are true
 - e. Only C is true
10. You recently diagnosed Sally who is 25 with T2DM. When would you start annual screening for microalbuminuria?
 - a. Five years after diagnosis
 - b. Screening should start at diagnosis
 - c. When she is 30 years old
 - d. No screening is required
11. Which of the following is a benefit of SMBG (self-monitoring of blood glucose) ?
 - a. Allows patients to determine if they're hitting their glycemic control target
 - b. SMBG is needed to prevent hypoglycemia
 - c. It can guide decisions by clinicians
 - d. Only A and B are true
 - e. A, B and C are true
12. Which of the following is TRUE regarding insulin needs in pregnancy?
 - a. Insulin needs remain constant throughout pregnancy
 - b. Insulin needs drop in the first trimester before rising again in the second and third trimesters
 - c. Insulin needs rise in the first and second trimesters before dropping in the third trimester
 - d. None of the above

13. A pregnant patient presents to your family medicine office at 25 weeks gestation for a routine visit. The patient's random blood glucose is 12 mM after drinking 50g of glucose. What is the most appropriate next step?
- Order a confirmatory HbA1c
 - Ask the patient to fast and then repeat the test with 75g of oral glucose instead of 50g
 - The patient has gestational diabetes (GDM). You should refer them to diabetes education and a dietitian.
 - None of the above
14. Which of the following is TRUE regarding diabetes and pregnancy?
- Both gestational diabetes (GDM) and pre-existing diabetes put the fetus at the risk of congenital malformations if uncontrolled.
 - GDM is not associated with a risk of congenital malformations, but it puts the pregnant patient at risk of premature birth, C-section and macrosomia among other things
 - GDM is associated with an increased risk of T2DM in the mother later in life.
 - Both B and C are true
 - None of the above
15. How are long-acting insulin preparations able to have a long duration of action?
- Due to their iron-binding properties
 - Due to first-pass metabolism in the liver
 - Due to their slow release from the hexameric form to the monomeric form
 - None of the above
16. Which of the following anti-hyperglycemic drugs is associated with the highest risk of hypoglycemia?
- Metformin
 - Sulfonylureas
 - SGLT2 inhibitors
 - GLP1-RAs
 - Both A and C
17. A 65-year-old patient with T2DM presents to the ER with dehydration and a significantly elevated creatinine. Which of the following medications might have contributed to this patient's symptoms?
- Metformin
 - SGLT2 inhibitor
 - GLP-RA
 - Insulin
18. A 30-year-old patient presents to your family medicine office with concerns regarding their weight. The patient meets the criteria for pharmacotherapy. As part of your general assessment, you run a patient health questionnaire (PHQ-9) and suspect that the patient might be suffering from depression. Which medication would you not prescribe to this patient?
- Orlistat
 - Semaglutide
 - Liraglutide

- d. Bupropion/Naltrexone
- 19. Which of the following is TRUE regarding bariatric surgery for obesity management?
 - a. Sleeve gastrectomy reduces the stomach to 25% of its original size
 - b. Sleeve gastrectomy is often recommended for patients with a high preoperative risk
 - c. Lapgastric band is not routinely recommended
 - d. All of the above are true

Answers

1. D. Both GAD and Zn18 are antibodies associated with T1DM
2. A. GLP-1 decreases in patients with T2DM. GIP increases in patients with T2DM but they don't produce insulin in response to the increase in GIP
3. B. A 4-dose MDI consisting of 3 fast-acting insulin doses before meals and a single long-acting would be best for reducing the risk of hypoglycemia in this patient. For example, if they have to skip a meal because of work, they can skip the insulin dose associated with that meal
4. B. A patient who has symptoms of diabetes (such as polyuria and polydipsia) and a random BG of 11.1 can be diagnosed with diabetes. No confirmatory test is required in this case
5. C. Patients with T2DM do not have an absolute insulin deficiency but they have insulin resistance. T2DM is less likely to present as an acute condition than T1DM. Patients with T2DM often have other features associated with metabolic syndrome
6. C. Having a first degree relative with T2DM increases the risk of T2DM. Monozygotic twins have a 90% risk of developing T2DM
7. C. Given the patient's symptoms, very high BG, normal pH and their family history of T2DM, HHS is the most likely diagnosis. There is nothing in the question that supports a diagnosis of a septic shock.
8. D. The presence of new blood vessels is indicative of proliferative retinopathy which requires treatment.
9. D. Refer to Dr. McManus's module on Prevention of Diabetes Complications
10. B. According to Diabetes Canada guidelines, screening for proteinuria begins at diagnosis for patients with T2DM and 5 years after diagnosis for patients with T1DM. Sally has T2DM so you would start screening her at diagnosis. (Link: <https://guidelines.diabetes.ca/cpg/chapter29#sec5>)
11. E. Refer to the targets for glycemic control module
12. B. Insulin needs drop in the first trimester before rising again slowly in the second trimester and rising more rapidly in the third trimester
13. C. The patient's BG is >7.8 mM/L after a 50g oral glucose tolerance test. This is sufficient for a diagnosis of GDM. No fasting 75g oral glucose tolerance test is required in this case
14. D. GDM is not associated with a risk of congenital malformations but it puts mother and fetus at risk of pregnancy complications
15. C. Long-acting insulin preparations are designed to dissociate slowly from the stable hexameric form.
16. B. Of the drugs listed here, SU's have the highest risk of hypoglycemia

17. B. Given the patient's symptoms (dehydration, renal failure..etc), it's most likely that they've had polyuria and hypovolemia due to SGLT2 inhibitor use.
18. D. Bupropion/Naltrexone may worsen depression initially
19. D. Refer to the Intro to Obesity module

Week 2

Questions

1. Which of the statement/s is/are true regarding abnormalities of the endocrine system (could be multiple correct answers)?
 - a. An ACTH stimulation is used to screen for suspected Cushing's syndrome
 - b. The most common cause of glucocorticoid excess is iatrogenic
 - c. The most common cause of thyrotoxicosis is iatrogenic
 - d. Hyperpigmentation and vitiligo are common presentations of Con's disease
2. Which of the following is consistent with thyroid mass complications?
 - a. Dysphonia
 - b. Exertional dyspnea
 - c. Dysphagia
 - d. All of the above
3. The prescription of ergocalciferol should be avoided or prescribed with caution in which of the following patient/s (could be multiple correct answers)?
 - a. A patient with hyperparathyroidism due to renal insufficiency
 - b. A patient taking thiazide diuretics
 - c. A patient taking furosemide
 - d. A patient with hyperphosphatemia
4. A patient with hypocalcemia is prescribed a vitamin D supplement. They inquire about the effects of vitamin D on their bone health. How would you counsel the patient?
 - a. Since their serum calcium is currently low, vitamin D will promote bone resorption.
 - b. Since their serum calcium is currently low, vitamin D will promote bone formation.
 - c. Vitamin D will promote bone resorption regardless of their serum calcium levels.
 - d. Vitamin D will promote bone formation regardless of their serum calcium levels.
5. All of the following could possibly present in a patient with hypothyroidism except:
 - a. Decreased basal metabolic activity
 - b. Excessive lipolysis, glycolysis and gluconeogenesis
 - c. Defect in early neural development
 - d. Respiratory disorders concerning the lungs
6. What would be a consequence of malfunctioning chief cells in the parathyroid gland?
 - a. Decreased levels of parathyroid hormones despite normal or low calcium levels.
 - b. Desensitization of parathyroid hormone receptors in the intestines.
 - c. Increased calcium absorption in the intestines despite normal or low calcium levels
 - d. Increased calcium reabsorption in the kidneys despite normal or high calcium levels.
7. A patient presents with decreased heart rate, cold intolerance, and weight gain despite decreased appetite. Their lab results demonstrate high TSH and low FT4. You inquire the patient about past thyroid diseases and thyroid treatments (radiation, surgery), which are negative. You also screened the patient for risk of pituitary disease, which is also negative. What is your next appropriate step?

- a. Order FT3 level
 - b. Order anti-TPO and anti-Tg antibody levels
 - c. Order a radioactive iodine uptake test
 - d. Make the diagnosis of Hashimoto's disease
8. Which of the following will is not an appropriate treatment for each of the patients
- a. Using PTU as a first line treatment for Grave's disease in a pregnant patient
 - b. Using radioactive iodine to treat a patient with thyroid adenoma
 - c. Using radioactive iodine to treat a patient with toxic multinodular goiter
 - d. Using methimazole to treat a large symptomatic goiter
9. A patient presents with a 1.5 cm palpable thyroid mass on the right. You measured their TSH and it was normal. What is your next step?
- a. Perform a thyroid ultrasound
 - b. Perform a radioactive iodine uptake test
 - c. Order T4 and T4 levels
 - d. Perform a fine needle aspiration biopsy
10. Which of the following thyroid masses require ultrasound surveillance
- a. Low risk ultrasound result with benign cytology
 - b. Moderate ultrasound result with benign cytology
 - c. High risk ultrasound result with benign cytology
 - d. High risk ultrasound result with malignant cytology
11. A patient presents acutely with perioral paresthesia, carpo-pedal spasm and palpitations. Their correct serum calcium is 1.8 mmol/l. Which of the following would be the most appropriate to administer in this acute setting.
- a. Calcitriol
 - b. Furosemide
 - c. Thiazide diuretic
 - d. Calcium gluconate
12. Which one of the following are possible symptoms/signs of hypercalcemia (could be multiple correct answers)
- a. Abdominal distress
 - b. Shortened QT interval
 - c. Apathy
 - d. Spontaneous fractures
 - e. Kidney stones
13. Which of the following warrant consideration of pharmacotherapy for osteoporosis in patients with moderate risk of fractures (could be multiple possible answers)?
- a. Recurrent falls
 - b. Parental history of vertebral fractures
 - c. Aromatase therapy for breast cancer
 - d. Smoking
 - e. Androgen deprivation therapy for prostate cancer

14. Your patient with osteoporosis and a previous hip fracture has been treated with romosozumab for 6 months. Their bone mineral density has been stable and they experienced no fracture events so far. How would you continue the management?
- Romosozumab seem to be working well for the patient, so you will keep them on romosozumab until their osteoporosis progress or they experience a fracture
 - Add teriparatide to the treatment
 - Follow up with zoledronic acid after 1 year of romosozumab therapy.
 - Follow up with raloxifene after 1 year of romosozumab therapy.
15. Which of the following is true regarding bone diseases?
- Paget's disease involves a defect in bone mineralization while osteomalacia involve a defect in bone remodeling
 - Cotton wool appearance is a clinical presentation of osteogenesis imperfecta
 - Teriparatide can be used to treat Paget's disease.
 - Hearing loss can be a clinical presentation for both Paget's disease and osteogenesis imperfecta.
16. A patient presents with decreased serum calcium and phosphates, with increased alkaline phosphatase. Upon physical exam, you also discover proximal muscle weakness. You observe pseudofractures in the left femoral neck on radiography. Which of the following bone diseases does this patient most likely have.
- Paget's disease
 - Osteoporosis
 - Osteogenesis imperfecta
 - Osteomalacia
17. A baby was born with anencephaly. During which week of embryogenesis did this defect most likely occur.
- Week 1
 - Week 2
 - Week 3
 - Week 4
18. If a teratogen is introduced during week 6 of embryogenesis, which of the following will be the most likely effect on the developing embryo.
- The teratogen either kills the embryo or has no effect on it
 - The teratogen could induce major congenital heart abnormalities
 - The teratogen could induce minor morphological abnormalities
 - The teratogen could induce minor functional abnormalities.
19. How does GnRH regulate gonadotropin release?
- The hypothalamus secretes GnRH into the general circulation and acts on the receptors of the posterior pituitary to induce the secretion of gonadotropins.
 - The hypothalamus secretes GnRH into the general circulation and acts on the receptors of the anterior pituitary to induce the secretion of gonadotropins.
 - The hypothalamus secretes GnRH through a portal system that directly reaches the receptors on the posterior pituitary and induces secretion of gonadotropins.

- d. The hypothalamus secretes GnRH through a portal system that directly reaches the receptors on the anterior pituitary and induces the secretion of gonadotropins.
- 20. Which of the following is FALSE regarding the regulation of the GnRH pulse generator
 - a. GABA is the strongest positive regulator of the GnRH generator
 - b. The pulsatile release of GnRH is to prevent receptor desensitization
 - c. Steroids indirectly regulate GnRH release by acting on neurons that synapse onto the GnRH neurons.
 - d. The pulse generator capacity of the GnRH neurons is intrinsic.
- 21. For a 25 year old female who has a regular menstrual cycle, which of the following is correct
 - a. We can't be certain of the duration of her luteal phase since it varies across women, but her follicular phase is likely 14 days.
 - b. Her FSH levels are the lowest after she ovulates.
 - c. Her estrogen levels will continuously decrease following ovulation
 - d. There is one developing follicle in her ovaries during each menstrual cycle
- 22. What induces the mid-cycle LH-FSH surge
 - a. When estrogen reaches and maintains a high enough level
 - b. When estrogen levels drop and no longer exerts negative feedback on the HPO axis
 - c. When GnRH pulses are the strongest
 - d. When the corpus luteum begins to produce progesterone.
- 23. Which of the following is correct regarding oocyte development
 - a. The oocyte undergoes both mitosis and meiosis in a women of reproductive age
 - b. There is an even division of DNA and cellular content upon completion of meiosis I
 - c. Following ovulation, the oocyte rests in metaphase of meiosis I and transits through the fallopian tube, awaiting fertilization.
 - d. Nondisjunction in oocytes can occur upon fertilization with a sperm.
- 24. What are the hormones which granulosa cells and theca cells respond to, respectively?
 - a. Granulosa cells response to estrogen and theca cells respond to progesterone
 - b. Granulosa cells respond to FSH and theca cells respond to LH
 - c. Both granulosa cells and theca cells respond to GnRH
 - d. Granulosa cells respond to progesterone and theca cells respond to estrogen

Answers

1. B, C. A is false because ACTH stimulation is used to screen for suspected adrenal failure. D is false because hyperpigmentation and vitiligo are common presentations of Addison's disease (primary adrenal insufficiency).
2. D. Exertional dyspnea, dysphagia, and dysphonia is due to compression of the trachea, esophagus, and recurrent laryngeal nerve, respectively.
3. A, B, D. Furosemide promotes calcium excretion, thus it is helpful in preventing hypercalcemia when taking vitamin D. Ergocalciferol requires both hepatic and renal metabolism to become the active calcitriol. A patient with renal insufficiency can convert ergocalciferol into calcifediol (hepatic), but can't effectively convert calcifediol into calcitriol (renal). Thiazide increases calcium

reabsorption, thus it should be taken with caution in conjunction with vitamin D to prevent hypercalcemia. Vitamin D increases both calcium and phosphate absorption, thus can worsen hypophosphatemia.

4. A.
5. B. A lack of thyroid hormones will decrease lipolysis, glycolysis and gluconeogenesis. Thyroid hormones are necessary for maintaining basal metabolic activity, permissive to GH action and necessary for induction of surfactant.
6. D. The chief cells are responsible for responding to high levels of calcium by shutting down parathyroid hormone release. If the chief cells malfunction, they will fail to detect hypercalcemia and the parathyroid hormone levels will remain high, which induces the kidneys to reabsorb even more calcium.
7. D. If you have excluded past thyroid disease and treatment and pituitary disease as causes of hypothyroidism, you can assume the diagnosis of Hashimoto's disease based on the elevated FSH and low FT4 level. You do not need to order FT3 for assessing hypothyroidism. Radioactive iodine uptake test is ordered to assess causes for low TSH levels. Anti-TPO and anti-Tg antibodies are sensitive for Hashimoto's disease, but they almost never influence clinical decisions in hypothyroidism, so they don't need to be ordered.
8. D. A large symptomatic goiter should be treated with surgery. Radioactive iodine scan and uptake is the first line treatment for thyroid adenoma and toxic multinodular goitre. PTU is less teratogenic compared to methimazole, thus is used to treat Grave's disease in pregnant patients.
9. A. When investigating a thyroid mass with normal TSH level, the next step is to perform an ultrasound and obtain a TIRADS score. The TIRADS score along with the mass size will decipher whether a biopsy is necessary.
10. B. A should be followed clinically. C should be followed with repeat biopsy. D should be treated with surgery.
11. D. Calcium gluconate should be used to treat acute hypocalcemia. Calcitriol and thiazide diuretics are used to manage chronic hypocalcemia. Furosemide promotes the excretion of calcium, thus should be avoided in hypocalcemia.
12. ABCDE. These are all possible signs and symptoms of hypercalcemia. The shortened QT interval in hypercalcemia can lead to arrhythmias. Remember "stones (kidney), bones, groans (GI), and psychiatric overtones" for hypercalcemia.
13. A, C, E.
14. C. Romosozumab, a bone formation therapy, should be followed with anti-resorptive therapy after 1 year. Both zoledronic acid and raloxifene are anti-resorptive therapies, but raloxifene is not effective for hip fractures.
15. D. Skull abnormalities in Paget's disease and osteogenesis imperfecta can lead to compression and damage of cranial nerves, resulting in hearing loss. Osteomalacia involves a defect in bone mineralization while Paget's disease involves a defect in bone remodeling. Cotton wool appearance is seen in Paget's disease, representing increased and abnormal bone remodeling. Bone formation therapies, such as teriparatide, should be avoided in Paget's disease since there is already increased bone formation.
16. D.

17. D. Neurulation is completed in week 4, during which the caudal and rostral ends. Failure of the caudal end to close results in anencephaly.
18. B. Week 3-8 is the most critical period for cell differentiation, organogenesis, and morphogenesis, thus a teratogen exposure at week 6 can potentially introduce a major congenital abnormality. Teratogens have an all or nothing effect during the first 2 weeks of development. Teratogens introduced during the fetal period (week 9 onwards) could potentially introduce minor morphological and functional abnormalities.
19. D.
20. A. Kisspeptin is the strongest positive regulator, GABA is a negative regulator.
21. B. FSH levels are at the lowest after ovulation due to negative feedback from estrogen and progesterone. The length of the follicular phase varies across women, but the luteal phase is usually consistently 14 days. The estrogen level has another surge in the luteal phase. There are multiple follicular waves in development during each cycle.
22. A.
23. D. Upon fertilization, the oocyte undergoes completion of meiosis II, where nondisjunction can occur. Oocytes only undergo mitosis during gestation, women are born with a set number of oocytes. There is an even division of DNA with meiosis I, but the first polar body contains very little oocyte cytoplasm. Following ovulation, the oocyte rests in metaphase of meiosis II.
24. B

Week 3

Questions

1. Which of the following patients requires the most urgent management:
 - a. Archibald presents with weakness, anorexia, and salt cravings
 - b. Bastien presents with treatment resistant hypertension and hypokalemia
 - c. Christian presents with dehydration, shock, and unexplained hypoglycemia
 - d. Damien presents with atrophied peripheral muscles, large purple striae, and ecchymoses
2. What is the most common cause of Cushing's syndrome?
 - a. Exogenous cortisol
 - b. Pituitary adenoma
 - c. Adrenal adenoma
 - d. ACTH producing tumour
3. Which of the following statements describes an accurate relationship between the various hormones:
 - a. Somatostatin inhibits prolactin production
 - b. Gonadotropin releasing hormone stimulates growth hormone production
 - c. Dopamine inhibits thyroid stimulating hormone production
 - d. Thyroid releasing hormone stimulates prolactin production
4. What is the front line treatment for hyperprolactinemia?
 - a. Prolactin receptor antagonists
 - b. Dopamine receptor agonists
 - c. Surgery
 - d. Oral contraceptives
5. What is the difference between premenstrual syndrome (PMS) and premenstrual dysphoric disorder (PMDD)?
 - a. PMDD includes both physical and behavioural symptoms while PMS is purely behavioural
 - b. PMS occurs prior to the onset of PMDD
 - c. PMDD is more disruptive to daily functioning than PMS
 - d. The two are synonymous
6. Which of the following statements is FALSE?
 - a. The onset of menarche does not necessarily mean the full reproduction capacity has been reached
 - b. Adrenarche is the onset of axillary and pubic hair due to increased androgen production by the gonads
 - c. Gonadarche is the activation of gonads by FSH and LH
 - d. Thelarche is the appearance of breast tissue due to estrogen production by the ovaries
7. Correctly order the following pubertal milestones for girls from earliest to latest onset:
 - a. Growth spurt

- b. Menarche
 - c. Thelarche
 - d. Pubarche
8. Which of the following is NOT an accepted definition of delayed puberty?
- a. Absence of puberty >13 years old for females
 - b. Absence of puberty >15 years old for males
 - c. Absence of puberty > 2.5 SD of the average age for either sex
 - d. Incomplete puberty > 2.5 SD of the average age for either sex
9. When examining a 14 year old boy, which of the following findings would make you most concerned about abnormal growth?
- a. His height is >2 SD below the average for his age and sex
 - b. An X ray of wrists shows delayed bone age despite his short stature
 - c. His puberty has been delayed despite his short stature
 - d. He is currently significantly below his genetic potential
10. Anh is a 13 year old Vietnamese male who presents to your clinic with concerns of short stature. His mother's height is 155cm and his father's height is 171cm. What is his genetic height potential?
- a. 168cm
 - b. 169cm
 - c. 170cm
 - d. 171cm
11. Why is FSH generally lower in males than in females?
- a. FSH is not lower in males than in females
 - b. Follicular cells in females produce hormones that upregulate the secretion of FSH
 - c. Seminiferous cells in males produce hormones that downregulate the secretion of FSH
 - d. The mechanism is not currently known
12. What is the difference between spermatogenesis and spermiogenesis?
- a. The two are synonymous
 - b. Spermatogenesis produces spermatogonia while spermiogenesis produces spermatozoa
 - c. Spermatogenesis produces spermatocytes while spermiogenesis produces spermatids
 - d. Spermatogenesis produces spermatids while spermiogenesis produces spermatozoa
13. What is the primary driver of female external genitalia development?
- a. Presence of estrogen
 - b. Absence of DHT
 - c. Absence of testosterone
 - d. A and B
 - e. A and C
14. Which of the following structures degenerate in males over the course of development?
- a. Mesonephric duct
 - b. Paramesonephric duct
 - c. Gubernaculum
 - d. Cortical cords

15. Is an estrogen block required for female-to-male transition?
- a. Yes, so that the effects of testosterone is not counteracted
 - b. Yes, so that the feminizing effects of estrogen is downregulated
 - c. No, because testosterone will automatically downregulate estrogen
 - d. No, because the effects of testosterone supercedes the effect of estrogen
16. Risks of feminizing hormones include (select all that apply):
- a. Thromboembolic events
 - b. Gallstones
 - c. Weight gain
 - d. Hypertriglyceridemia
 - e. Elevated liver enzymes
 - f. Polycythemia
 - g. Sleep apnea
17. The primary laboratory result indicative of congenital adrenal hyperplasia is:
- a. Low serum 21 hydroxylase
 - b. High serum 17 hydroxyprogesterone
 - c. Low serum 11b hydroxylase
 - d. High serum androgen
18. Which of the following is NOT a potential cause of 46XY, DSD?
- a. Congenital adrenal hyperplasia
 - b. Testicular regression
 - c. 5-alpha reductase deficiency
 - d. Incomplete androgen insensitivity
19. The uterine artery can be found embedded in which endopelvic ligament?
- a. Transverse cervical ligament
 - b. Uterosacral ligament
 - c. Broad ligament
 - d. Suspensory ligament of the ovary
20. Match the structures to the lymph nodes that drain them (one answer per choice):
- i. Skin
 - ii. Erectile bodies
 - iii. Cervix
 - iv. Cavernous bodies
 - v. Inferior vagina
 - vi. Uterine fundus
- a. Sacral lymph nodes
 - b. External iliac lymph nodes
 - c. Superficial inguinal lymph nodes
 - d. Deep inguinal lymph nodes
 - e. Lumbar lymph nodes
 - f. Internal iliac lymph nodes

Answers

1. C. Christian presents with the classic signs of an adrenal crisis, which is a medical emergency.
2. A. While all the options are potential causes of Cushing's, exogenous cortisol/glucocorticoids is by far the most common cause.
3. D. Somatostatin inhibits TSH and GH, GnRH stimulates FSH and LH, dopamine inhibits prolactin, and TRH stimulates both TSH and prolactin.
4. B. Surgery is only with failed medical therapy. The other options are not indicated for hyperprolactinemia.
5. C. Both PMS and PMDD can have behavioural and physical symptoms. PMS may predicate PMDD but this is not required. PMDD is a much more disabling condition than PMS.
6. B. Adrenarche is due to the production of *adrenal* androgens, not from the gonads.
7. C, A, D, B.
8. B. The standard definition for males is >14 years old.
9. D. While having a height >2 SD below the average could be an indicator for abnormal growth, having short stature alone is not enough to indicate abnormal growth; he could just be on the shorter end of normal. Delayed bone age and delayed puberty indicate normal growth variants in individuals with shorter stature.
10. B. The equation for the genetic potential of males is $(\text{height of mother} + \text{height of father} + 13)/2$.
11. C. Sertoli cells in the seminiferous tubules produce inhibin, which downregulates FSH, as LH is the main hormone of spermatogenesis.
12. D.
13. D. DHT is more potent than testosterone and therefore a more critical influencer of sexual development.
14. B. The paramesonephric duct, also known as the Müllerian ducts give rise predominantly to female reproductive tract organs.
15. C.
16. A, B, C, D, E. Polycythemia and sleep apnea are effects of masculinizing hormones.
17. B. While the other statements can also be true of CAH, the primary diagnostic test is serum 17 hydroxyprogesterone.
18. A. CAH would be a common cause of 46XX, DSD (over-virilized female), not 46XY, DSD (under-virilized male).
19. A. The transverse cervical ligament, aka the cardinal ligament, contains both the uterine vessels and the ureters. The suspensory ligament of the ovary is not an endopelvic ligament, and contains the ovarian vessels.
20. A - v, B - iii, C - i, D - ii, E - vi, F - iv. Note that both the internal and external iliac lymph nodes could drain the cervix.

Week 4

Questions

1. Which of the following characteristic and corresponding phase of the sexual response cycle is INCORRECTLY matched.
 - a. Excitement – penile erection in males; breast enlargement and nipple erection in females
 - b. Plateau – increase in circumference in coronal ridge of penis in males, uterus becomes fully elevated in females
 - c. Organism – Contraction of the vas deferens, seminal vesicles and prostate in males; rhythmic contractions of vaginal walls in females
 - d. Resolution – males do not experience a refractory period and are able to ejaculate many times; females experience a refractory period making them incapable of experiencing multiple orgasms

2. Which of the following statements is true?
 - a. Hormones such as GABA and vasopressin have an excitatory effect in initiating the sexual response.
 - b. Orgasms are primarily through to be a cerebral event as it has been demonstrated in studies that those with spinal cord injuries can still experience orgasms.
 - c. During the plateau phase of the sexual response cycle, in females, blood pressure decreases in order to promote vasocongestion of the of the lower 1/3rd of the vagina.
 - d. Visual and tactile stimuli play a more important role in females compared to males in intimating the sexual response.

3. Which of the following is INCORRECT with regards to the presentation of a benign breast pathology?
 - a. Fat necrosis – Bloody nipple discharge
 - b. Fibroadenoma – Usually occur in women ages 20-30
 - c. Mastitis – tense, hot, painful breast
 - d. Fibrocystic change – masses that fluctuate throughout menstrual cycle

4. Which of the following is NOT a risk factor for breast cancer?
 - a. Early menarche
 - b. Alcohol
 - c. Multiparity
 - d. Does not breast feed

5. A 38yr old woman presents to the clinic with a palpable round lump in her left breast measuring 2.5 x 1.5 cm. The lump has irregular borders, is fixed, non-tender, and hard. She denies any chest pain, SOB, or systemic symptoms. Her past medical hx is significant for hypothyroidism, for which she takes levothyroxine. She does not smoke or drink. Physical exam revealed no other concerning findings. Which is the next best investigation for this patient?
 - A. Ultrasound
 - B. Mammogram
 - C. MRI
 - D. No investigations necessary at this time. Follow up in 3 months.

6. Which of the following statements about breast pain & its various causes are false?
 - a. Non-cyclical causes of breast pain in conditions such mastitis, medication (HRT/OCP), and Cooper's ligament stretch.
 - b. Supportive therapy for breast pain can include NSAIDs, hot/cold compresses, and supportive bras
 - c. Individuals with mastitis should refrain from breast feeding as there is a risk of infecting the infant with contaminated breast milk
 - d. The most common cause of mastitis is *Staphylococcus aureus*

7. A 43yr old women with irregular menstrual cycles presents for postcoital vaginal bleeding. As part of your work up, you order a pap smear. The results from the smear read – high grade squamous intraepithelial lesion (HSIL). What is the next best step in the management of this patient?
 - a. Refer to colposcopy and biopsy.
 - b. HPV DNA testing
 - c. Repeat cytology in 12 months
 - d. Repeat cytology immediately

8. Which of the following women are eligible for cervical cancer screening?
 - a. Woman A who is 25 and sexually active
 - b. Women B who is 23 and sexually active
 - c. Women C who is 21, immunocompromised, and sexually active.
 - d. All of the above
 - e. A and C

9. Which of the following infections are diagnosed using serology?
 - i. Chlamydia
 - ii. Gonorrhea
 - iii. Syphilis
 - iv. Hepatitis B
 - v. HIV
 - A. I, ii, IV
 - B. I, iii, V
 - C. III, I

D. III, IV, V

10. Which of the following symptoms is present during the secondary stage of syphilis infection?
 - a. Painless chancre
 - b. Condyloma lata
 - c. Gummatous syphilis
 - d. CNS complications (meningitis, stroke, or seizures)
11. What is the most common STI in the London/Middlesex region?
 - a. Chlamydia
 - b. Hepatitis C
 - c. Gonorrhea
 - d. HIV
12. Which of the following is not a risk factor for contracting an STI?
 - a. Injection drug use
 - b. Sexually active youth under 25 years of age.
 - c. No contraception or sole use of non-barrier methods of
 - d. Previous STI
 - e. All of the above are risk factors for STIs
13. Which one of the following is not part of the BRATS acronym for common genital injuries sustained from an acute sexual assault?
 - a. B – bruises
 - b. R – redness
 - c. A – abrasions
 - d. T – tears
 - e. S – soreness
14. You are working as a new resident doctor in the Sexual Assault and Domestic Violence Program at St. Joseph's hospital. You are assisting in seeing a 26 yr old female patient who was just recently sexually assaulted by an unknown assailant. She is quite distraught, emotional, and frightened. What would be the first step in your assessment of this patient?
 - a. Refer the patient to appropriate mental health professionals and social support groups
 - b. Prescribe the morning after pill (levonorgestrel), anti-retroviral medication for HIV prophylaxis, and antibiotics for prophylaxis of chlamydia and gonorrhea.
 - c. Perform a detailed history and physical exam, ensuring to gather medically relevant details about the assault and look for corroborating evidence on the physical.
 - d. Obtain informed consent from the patient to begin the examination, knowing that the patient may consent to any or all 3 of aspects of the exam: medical exam & tx for the effects of the assault; collection and documentation of medical legal evidence; and disclosure of results to the police for potential legal action.
15. A 26-year-old female presents to your clinic for complaints of foul and "fishy" smelling vaginal discharge that is grey in colour. Her medications only consist of a Mirena IUD. You perform some investigations and here are the results: Wet mount reveals 22% clue cells, pH of discharge is 5.7, and a positive Whiff test. What is the most likely diagnosis?

- a. Trichomonas
 - b. Bacterial vaginosis
 - c. Atrophic vaginitis
 - d. Candidiasis
16. Which of the following statements is not a difference between bacterial vaginosis (BV) and candidiasis?
- a. Candidiasis is caused by a fungal agent while BV is caused by a bacterial agent
 - b. Candidiasis is treated using metronidazole while BV is treated using Miconazole or Fluconazole
 - c. Candidiasis will show budding yeast, hyphae or pseudohyphae on wet mount while BV will show increased clue cells on wet mount
 - d. Candidiasis will present with cottage cheese like discharge while BV will present with thin white/grey vaginal discharge

Answers

1. D – this answer is reversed. Men experience a refractory period while women do not.
2. B - A is wrong because GABA and vasopressin are inhibitory. C is wrong because BP increases in plateau phase. D is wrong as visual/tactile stimuli are more important in men than women in initiating of sexual response cycle.
3. A - Intraductal papillomas typically present with bloody discharge, not fat necrosis.
4. C - Nulliparity is a risk factor for breast cancer, not multiparity.
5. B - The lump has malignant features, so you want to investigate it (D is eliminated). Mammogram is the ix modality of choice in women >35, so this eliminates A and C. It is possible that you would both do both a mammogram and an US for women over 35 but this question is asking what the next best investigation is, so it requires you that a mammogram is definitely needed for someone over 35 presenting with a breast lump with high suspicion for malignancy.
6. C - Individuals with mastitis should continue to breast feed as emptying the breast can help resolve the condition.
7. A - HSIL is an immediate referral to colposcopy.
8. E. Current Ontario screening recommendations (2022) say that you should delay screening for women until age 25 and if they are sexually active. However, you can begin screening at age 21 if they are immunocompromised and are or ever have been sexually active. The current guidelines do not recommend screening earlier than age 25 even if they are sexually active. See the attached guidance document in the module

Ontario Cervical Screening Program (OCSPP)

Screening Recommendations Summary

January 2022

Current cytology-based recommendations for eligible participants with a cervix who have ever been sexually active	Initiation Age ≥ 25 is the preferred age of initiation. See below for guidance on change from starting at age ≥ 21 *	Screening interval Every 3 years with cytology if screening test is negative	Cessation Age 70 if person has had 3 negative cytology results in routine screening in the previous 10 years
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***Guidance during the change to human papillomavirus (HPV) testing**

Ontario Health (Cancer Care Ontario) is working with the Ministry of Health to implement HPV testing in cervical screening in Ontario. Until then, please continue to use cytology-based screening. Recommendations from the Canadian Task Force on Preventive Health Care and recent evidence support a higher age of initiation for cervical screening (with cytology or HPV testing). Primary care providers are now encouraged to initiate cytology-based screening at age 25 for those who are or have ever been sexually active, with one exception: people who are immunocompromised and are or have ever been sexually active can continue to start screening at age 21. Please note, primary care cancer screening tools and resources are not yet aligned with this guidance and will be updated with HPV implementation*.

9. D
10. B
11. A
12. E
13. E - The S in BRATS is swelling, not soreness.
14. D - All of these options may be included in your assessment of a sexual assault survivor.
However, the most important step is to obtain informed consent from the patient before you begin.
15. B
16. C

Week 5

Questions

1. Which of the following are not a characteristic of Progestin?
 - a. Many POCs increase aldosterone production, leading to fluid retention and weight gain.
 - b. POCs thicken cervical mucus, making it difficult for sperm penetration
 - c. The effect of Progesterone on GnRH suppression is weaker than Estrogen's effect
 - d. When taking POC, patient compliance is not as important
2. True or False: Taking antibiotics interferes with COCs effectiveness in contraception
 - a. True
 - b. False
3. Where does capacitation occur?
 - a. Fallopian tube
 - b. Uterus
 - c. Vagina
 - d. Cervix
4. True or false: egg deposited in cul-de-sac by transvaginal injection can never lead to conception.
 - a. True
 - b. False
5. Which of the following contraceptive methods should never be used in high risk individuals?
 - a. Diaphragm and Cervical cap
 - b. Nonoxynol-9
 - c. Tubal ligation
 - d. Condoms
6. Which of the following statements are FALSE?
 - a. The most common complication of tubal ligation is patient regret
 - b. Women must take contraceptives for maximum 1 month post procedure.
 - c. Vasectomy is safer than tubal ligation and hysteroscopic tubal occlusion
 - d. Progestin-only pills suppress ovulation in 50%
7. Which of the following are not offered methods of abortion in First Trimester?
 - a. Manual Vacuum Aspiration
 - b. Mifepristone + Misoprostol
 - c. Misoprostol + Methotrexate
 - d. Dilation and Curettage
 - e. Misoprostol + intracardiac injection of KCl or Digoxin

8. Which of the following are NOT a potential complication of D&C?
- Cervical shock
 - Uterine perforation
 - Post abortal syndrome (Hematometra)
 - Hemorrhage
 - Infertility
9. Which of the following are not common places for ectopic pregnancy?
- Fallopian tube
 - Ovaries
 - C-section scars
 - Urethra
10. Which of the following statements are FALSE?
- Ruptured ovarian cysts are commonly self-limited
 - Risk of ovarian torsion increases when there is ovarian enlargement
 - Ectopic pregnancy often leads to ovarian ischemia, resulting in ovarian necrosis, infarction and local hemorrhage
 - Ovarian torsion can occur in women of all ages
11. Dysmenorrhea is... (select all that applies)
- Purely a clinical diagnosis
 - There are no abnormal laboratory nor physical findings
 - Best treated with NSAIDS or ovulatory suppression (OCP)
 - Occurs due to prostaglandin release, which ultimately leads to stimulation of type A neurons.
12. True or False: first line treatment for endometriosis is hysterectomy and/or laparoscopic excision/ablation.
- True
 - False
13. Which of the following are not on the differential diagnosis for chronic pelvic pain?
- Dysmenorrhea
 - Endometriosis
 - Adenomyosis
 - Fibroids
 - Ruptured ovarian cyst
14. Which of the following are not considered hormone-dependent growth?
- Endometrial polyp
 - Adenomyosis

c. Leiomyoma

15. Which of the following is the most common benign neoplasm in females?

- a. Endometrial polyp
- b. Adenomyosis
- c. Leiomyoma
- d. Endometrial hyperplasia

16. True or False: Borderline tumor may progress to high grade carcinoma

- a. True
- b. False

17. Malignant epithelial tumors are... (select all that applies)

- a. Serous carcinoma is #1 form of ovarian cancer
- b. Often found unilaterally
- c. Often found at advanced stage
- d. Often post menopausal (younger if familial)

18. Match the correct marker substances for each pathology

- a. Dysgerminoma
- b. Yolk sac tumor
- c. Choriocarcinoma

I. b-hCG

II. a-FP

III. LDH

19. Which of the following conditions mimic PCOS?

- a. Exogenous testosterone intake
- b. Pregnancy
- c. Ovarian Tumor
- d. CAH
- e. All of the above

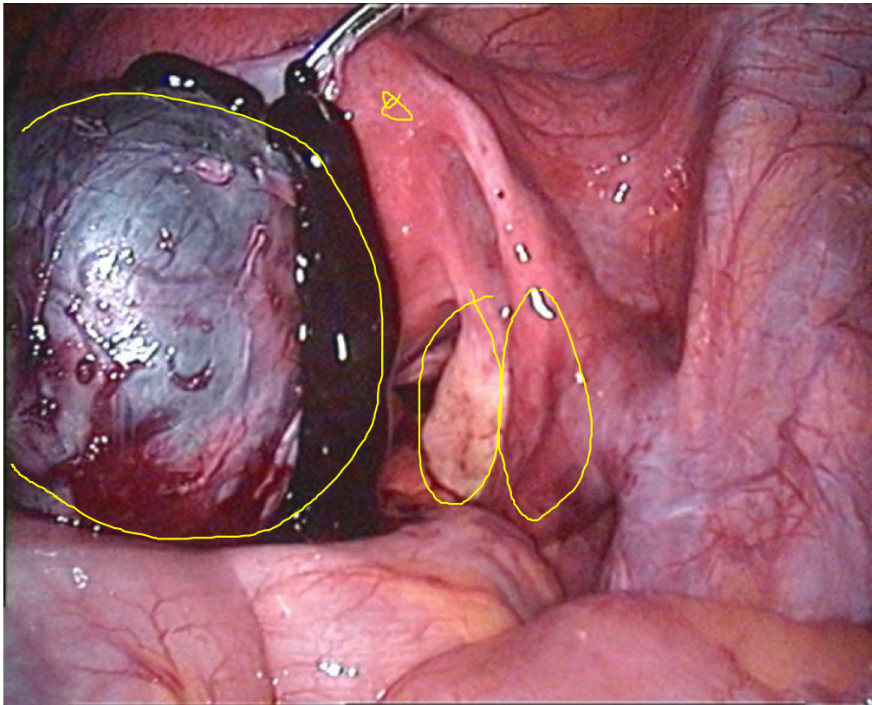
20. Select common features found in patients with PCOS

- a. Acanthosis
- b. Hirsutism
- c. Acne
- d. Female alopecia
- e. All of the above

21. Which of the following are recommended during workup of PCOS? Select all that applies.

- a. Prolactin levels
- b. Hemoglobin levels
- c. B-hCG
- d. HbA1c
- e. All of the above

22. What is this lesion called? What pathology is associated with this finding?



- a. Blueberry lesion - Leiomyoma
- b. Powder burn lesion - Leiomyoma
- c. Chocolate cyst - Endometriosis
- d. Peritoneal window - Endometriosis

23. Endometriosis can be treated with the following, except:

- a. NSAIDs
- b. Danazol
- c. NuvaRing
- d. Hysterectomy
- e. Tubal occlusion

24. Post Menopausal Bleeding is any bleeding in post-menopausal woman after absence of menstrual flow for:

- a. 3 consecutive months
- b. 2 years
- c. 5 months
- d. 12 months

25. Select an incorrect statement:
- a. Type 1 endometrial cancer is more common than Type 2 endometrial cancer
 - b. Endometrial carcinoma must be ruled out in patients presenting with PMB
 - c. Type 2 endometrial cancer is largely caused by unopposed estrogen
 - d. Conditions such as germ cell tumor and PCOS are sources of endogenous estrogen
26. First line treatment when endometrial pathology reports atypia endometrial hyperplasia?
- a. Hysterectomy with bilateral salpingo-oophorectomy
 - b. Medical management - Progestin-only
 - c. Oophorectomy only
 - d. Hysterectomy only
27. Which of the following occurs in menopause? Select all that applies.
- a. Hot flashes, vaginal atrophy, vaginal dryness are common symptoms
 - b. Increased vaginal hydration
 - c. Estrogen and progesterone production increases
 - d. LH and FSH secretion increases
28. Which of the following statements are true? Select all that apply.
- a. Post menopausal women with history of hysterectomy can be on Estrogen-only HRT.
 - b. LET alleviates vasomotor symptoms, such as palpitations and perspirations
 - c. The 3 main causes of illness and disability in developed countries for postmenopausal women are: Stroke, decreased sexual drive, and cancer
 - d. During perimenopause, women often experience change in their cycle length & flow
 - e. As women transition from perimenopause to menopause, they experience oligomenorrhea and hypomenorrhea, until menstruation completely stops
29. What do people usually mean when they describe "partial" hysterectomy?
- a. Hysterectomy only
 - b. Hysterectomy + Oophorectomy
 - c. Hysterectomy + Salpingectomy
 - d. Hysterectomy + cervix removal
 - e. Oophorectomy only
30. Which of the following procedure is recommended for patients with enlarged uterus?
- a. Abdominal hysterectomy
 - b. Vaginal hysterectomy
 - c. Laparoscopic-Assisted Vaginal hysterectomy (LAVH)
 - d. Total/subtotal laparoscopic hysterectomy

Answers

1. D. It is paramount that POCs are taken at the same time, every single day, because progesterone's effect on GnRH suppression is "weaker" as it only decreases the *chance* of ovulation suppression. It heavily relies on its secondary effects (i.e., thickening of cervical mucus, alteration of tubal peristalsis, suppression of uterine contractility), and therefore these effects wane over time, reverting the endometrium back to the "receptive" condition.
2. B. This is a common myth; there is no clinical relevance between antibiotic consumption & COC's effectiveness. However, if the patient develops side effect from antibiotics (i.e., vomiting, diarrhea), it may affect plasma concentration of OCPs, therefore making it "less effective".
3. A. Capacitation is the cellular and molecular changes that ejaculated spermatozoa must undergo in order to fertilize. It occurs while the sperm are in the fallopian tube. It is characterized by three accomplishments: 1) The ability to bind to the zona pellucida (noncellular porous layer of glycoproteins secreted by the oocyte; separates the oocyte from cumulus cells). 2) The ability to undergo acrosomal reaction. 3) The acquisition of hypermotility.
4. B. It can in fact be picked up by the fallopian tubes; in fact, some women were shown to achieve pregnancy despite having only one ovary and a single tube located on the contralateral side. The tubal transport is a complex system! It is dependent on 1) ciliary motion, 2) muscular contractility, and 3) tubal fluid → these contribute to varying degrees to effective tubal transport.
5. B. Nanoxynol-9 is a spermicide that causes vaginal epithelium damage and therefore increase risk of STI and HIV transmission. This should absolutely not be used in high risk individuals and in fact, has largely been removed from the market.
6. B. After hysteroscopic tubal ligation, it takes 3 months for tissue to grow to fill the inserted coil and therefore block/occlude the fallopian tube. Therefore, patients must take contraceptive methods for minimum of 3 months post operation - in fact, patients need 3 month follow up with imaging.
7. E. This option is used in 2nd trimester abortion.
8. E.
9. D. Ectopic pregnancy occurs in the fallopian tube most of the times. Other sites include: ovarian, cervical, cornual, abdominal, C-section scars.
10. C. Ovarian torsion often results in reduced venous outflow while there is continuous arterial inflow. This leads to ovarian edema → ovarian enlargement → further vascular compression → ovarian ischemia → ovarian necrosis, infarction and local hemorrhage.
11. A,B,C. Dysmenorrhea Occurs due to prostaglandin release, which ultimately leads to stimulation of type C neurons.
12. B. False: although the definitive management of endometriosis is hysterectomy with or without oophorectomy, this is not a first line treatment. Rather, the mainstay treatment of endometriosis is ovulatory suppression and pain control using NSAIDs.
13. E. Ruptured ovarian cyst would present as acute pelvic pain.
14. B.
15. C.

16. B. False: Borderline tumor may progress to low grade carcinoma.
17. B. Malignant Epithelial tumors are often bilateral; except mucinous.
18. a-iii/ b-ii/ c-i
19. E. Exogenous testosterone, CAH, and ovarian tumor lead to hyperandrogenism; while pregnancy causes oligomenorrhea. Hyperandrogenism & Oligomenorrhea are common features of PCOS.
20. E.
21. A, C, D. Should order **testosterone, prolactin, b-hcg, HbA1c & lipids** and sometimes 17 a-hydroxyprogesterone (to r/o CAH) and U/S (if history of oligomenorrhea nor hyperandrogenic labs are not present).
22. C. This substance leaking is old blood + endometrial tissue & often has appearance of chocolate fondue.
23. E.
24. D.
25. C. Type 2 is not related to unopposed estrogen exposure, while Type 1 is. Furthermore, Type 1 endometrial cancer is more common, found in younger aged patients than Type 2, and associated with familial history (i.e., Lynch syndrome, BRCA mutation).
26. A. With atypia, hysterectomy with bilateral salpingo-oophorectomy is recommended as it is associated with high risk of concurrent or subsequent malignancy. Only when surgery is contraindicated are patients recommended with high dose progesterone treatment. In contrast, typical endometrial hyperplasia is treated with Progesterone (i.e., LN-IUD) & should be followed up usually in 3 months with repeat biopsy.
27. A, D.
28. A, D, E.
29. A, C, D. "Partial hysterectomy" usually refers to removal of **uterus +/- cervix, +/- fallopian tube (salpingectomy)**, but with retention of ovaries (no oophorectomy)
30. A.

Week 6

Questions

1. Which of the following is true about hCG levels in a viable intrauterine pregnancy?
 - a. hCG levels increase exponentially until the end of the second trimester
 - b. hCG levels start to rise in the second trimester
 - c. hCG levels plateau during the late first trimester
 - d. hCG levels remain constant until delivery

2. Once an embryo is clearly seen on ultrasound, the most accurate estimate of gestational age can be provided by:
 - a. Crown to rump length (CRL)
 - b. Mean sac diameter (MSD)
 - c. Yolk sac
 - d. hCG levels

3. What is the most common cause of an ectopic pregnancy?
 - a. Endometriosis
 - b. IUD use
 - c. Pelvic inflammatory disease
 - d. Adenomyosis

4. Which of the following is true regarding gestational trophoblastic disease?
 - a. The most common site of metastases for choriocarcinoma is the liver
 - b. A snowstorm pattern on pelvic ultrasound without a fetus present suggests complete hydatidiform mole
 - c. Partial hydatidiform moles contain only paternal chromosomes
 - d. After a complete hydatidiform mole, patients should not conceive until at least 3 months after serum beta hCG has returned to baseline

5. In which case would you manage an ectopic pregnancy surgically instead of medically?
 - a. Patient is hemodynamically stable
 - b. Patient's mass has ruptured
 - c. Patient is not breastfeeding
 - d. Patient is compliant to follow-up

6. Which of the following about medical management for spontaneous abortion is false?
 - a. Mifepristone + misoprostol combination therapy is very costly
 - b. Mifepristone + misoprostol combination therapy is first line treatment for threatened spontaneous abortion
 - c. Side effects of misoprostol include nausea, diarrhea and fever
 - d. IV broad spectrum antibiotics should be used in septic spontaneous abortion

7. Which of the following are the 3 most common complications of D&C?
 - a. Infection
 - b. Uterine perforation
 - c. Hemorrhage
 - d. Hematometra
 - e. Asherman's Syndrome
 - f. Retained products of conception

8. A patient presents to your clinic following a D&C with acute pain and urinary retention but no vaginal bleeding. Which complication does she likely have?
 - a. Infection
 - b. Hematometra
 - c. Cervical shock
 - d. Cervical trauma

9. Which of the following are the most commonly required supplements in pregnancy
 - a. Iron and vitamin C
 - b. Folate and calcium
 - c. Zinc and folate
 - d. Iron and folate

10. Which of the following vaccinations is recommended in all pregnancies?
 - a. Pertussis (Tdap)
 - b. Measles-Mumps-Rubella (MMR)
 - c. Hepatitis A
 - d. BCC for tuberculosis

11. A new patient comes to your clinic, and you confirm he is 11 weeks pregnant. Which of the following is not a recommended investigation?
- Rubella test
 - Enhanced first trimester screening (eFTS)
 - 2-hour glucose tolerance test
 - Group and screen
12. Which of the following symptoms is more commonly experienced during the second and third trimesters, as opposed to the first trimester?
- Amenorrhea
 - Abdominal bloating
 - Mood swings
 - Breast tenderness
13. In which of the following cases would an OHIP-covered NIPT be indicated?
- Pregnant parent older than 35 years
 - Parents have a previous child with trisomy 21
 - Nuchal translucency > 3.0mm
 - Pregnant parent's first gestation
14. Which of the following is a risk of amniocentesis?
- Miscarriage risk of ~0.5%
 - Miscarriage risk of ~1%
 - Maternal cell contamination
 - Confined placental mosaicism
15. Which of the following pregnant people should be offered an ultrasound for fetal assessment?
- A 50-year-old G2T1L0 at 15 weeks gestation who had a stillbirth one year ago with a baby who was diagnosed with a complex congenital heart defect
 - A 41-year-old G2 in her second pregnancy at 8 weeks gestation whose previous child has Trisomy 21
 - A 30-year-old G6T0P0A5L0 with 5 previous first trimester pregnancy losses who is now 16 weeks gestation
 - An 18-year-old in her first pregnancy at 19 weeks with no significant risk factors identified at the initial prenatal visit
16. Which of the following is a part of the standard second trimester ultrasound?
- Assessment of fetal "well-being" through Biophysical Profile
 - Assessment of gestational sac diameter
 - Assessment of placental location
 - Assessment of amount of amniotic fluid

17. Which of the following teratogen exposures will pose the greatest risk to the fetus?
- A pregnant patient contracts influenza despite having received the influenza vaccine at 20 weeks gestation
 - A pregnant patient contracts Rubella at 25 weeks gestation
 - A pregnant patient starts smoking at 36 weeks gestation
 - A pregnant patient smokes from conception until they discover they're pregnant at 10 weeks gestation
18. Which of the following is not considered a teratogen?
- Zika virus
 - Toxoplasmosis
 - Hyperthermia
 - Acetaminophen
19. Which of the following is NOT a contraindication to labour?
- Active genital herpes
 - Placenta previa
 - Unfavourable pelvic architecture
 - Abnormal fetal lie
 - Significant prior uterine surgery
20. Prior to normal labour, which of the following occurs?
- The cervix becomes thicker and stronger to help push out the baby – this is known as cervical effacement
 - Braxton-Hicks contractions occur and indicate the onset of labour
 - The fetal heart rate begins to undergo regular decelerations
 - The fetal head becomes fixed in the pelvis – this is known as engagement
21. During the 1st stage of labour, which of the following is NOT a consideration?
- More frequent checks of cervical effacement and dilatation improve outcomes
 - A score of +2 for assessing fetal station indicates the head of the fetus is in the vagina
 - A fully dilated cervix is approximately 10 cm in diameter
 - A vaginal exam can assess fetal position, with an anterior occiput being the most favourable for delivery
 - A fetus with breech presentation may or may not be delivered vaginally
22. Which of the following regarding the 2nd stage of labour is FALSE?
- This process usually occurs more rapidly in multiparous women compared to primiparous women
 - Cardinal movements describe the series of movements the baby undergoes during delivery
 - Placental delivery usually occurs within the first 30 minutes

- D. A pelvic xray is not an adequate test to determine pelvic adequacy for delivery
23. Which of the following is NOT a contraindication for maternal discharge after delivery?
- A. Fever
 - B. Labour dystocia
 - C. Postpartum hemorrhage
 - D. Pre-eclampsia
 - E. Unresolved gestational hypertension
24. Which of the following is correct regarding postpartum care?
- A. Contraception is not required after birth, particularly if a woman is breastfeeding
 - B. Per rectum suppositories can help with postpartum hemorrhoids and constipation in 4th degree tears
 - C. An initial breastfeed of golden yellow milk is a common sign of mastitis
 - D. Sitz baths are recommended for perineal tears
25. Pam, one of your patients, successfully delivered a healthy baby girl, Cece, by spontaneous vaginal delivery 2 days ago. There were no complications and she is ready to be discharged. While her husband Jim searches for their car, she tells you that she already feels drained as she did not sleep well last night. Which of the following statements is NOT true?
- A. She may experience periods of weepiness and irritability in addition to her insomnia over the next few days, known as the “blues”
 - B. She can minimize her sleep disturbances from feedings throughout the night by reducing her exposure to bright lights
 - C. Having a baby is a significant life event and it is normal to experience an adjustment period that may affect her sleep
 - D. She should seek help if the “blues” persist beyond 2 weeks as she may have postpartum depression
 - E. After delivery, she will be in a hyperestrogenic state that may be contributing to her insomnia
26. Which of the following is NOT a responsibility that family physicians with extra training in obstetrical care provide?
- A. Provide antenatal, intrapartum, and postpartum care for low-risk pregnancies
 - B. Administer oxytocin as necessarily during labour
 - C. Provide care for patients with all degrees of postpartum hemorrhage
 - D. Perform vacuum-assisted vaginal deliveries
 - E. Provide routine maternal and neonatal care
27. Which of the following statements about midwifery is FALSE?
- A. A consultation with a midwife tends to run longer than with physicians, which allows them to provide more holistic care
 - B. Midwifery availability and regulation varies across Canada

- C. All midwives can provide support with medications like epidurals and oxytocin
 - D. The midwifery model requires midwives to provide informed choice rather than informed consent
 - E. One benefit to midwives is that they may provide home visits to patients
28. Which of the following regarding lactation and milk let-down regulation is true?
- A. Nipple stimulation inhibits the release of oxytocin
 - B. The posterior pituitary releases prolactin to facilitate milk let-down
 - C. The anterior pituitary releases oxytocin, which drives lactogenesis
 - D. Dopamine levels decrease in response to nipple stimulation and milk removal, which drives an increase in prolactin
 - E. Placental delivery drives an increase in progesterone, which facilitates lactogenesis
29. Rachel has recently become a mother to a healthy baby girl, Emma. Although the pregnancy was unexpected, her partner Ross is ecstatic, despite the 3% chance of pregnancy with their contraceptive of choice. Ross insists that Emma be breastfed, although Rachel is unsure. Which of the following statements regarding breastfeeding is FALSE and should not be used to help Rachel decide to breastfeed?
- A. Breastfeeding provides a host of long-term health benefits to both Emma and Rachel
 - B. Breastmilk is a sustainable, waste-free, environmentally-friendly practice
 - C. Breastmilk is the recommended exclusive nutrition for the first 6 months of life
 - D. Breastfeeding is a reliable method of contraception
 - E. Colostrum will provide Emma passive immunity to help prevent infections
 - F. Breastfeeding can help accelerate Rachel's recovery from childbirth
30. Which of the following contraceptives should NOT be used by a woman who is breastfeeding and in a heterosexual relationship to avoid future pregnancy?
- A. Depo-Provera
 - B. Progesterone-only pill
 - C. Condoms
 - D. Mirena IUD
 - E. Combined oral contraceptive
31. In the Newborn Resuscitation Program, what is the primary resuscitation intervention of focus?
- A. Oxygenation—Give free flow O₂
 - B. Circulation—perform CPR
 - C. Airway obstruction—always suction the airway
 - D. Ventilation—give positive pressure ventilation
 - E. Hypovolemia—give fluid bolus
32. Which of the following statements regarding normal cardiovascular and pulmonary adaptation is true?
- A. Surfactant is made in the lungs to increase surface tension within the alveoli
 - B. The foramen ovale closes after birth due to increased pulmonary vascular resistance
 - C. Lung fluid reabsorption in the neonate is facilitated by ENaC channels within the alveoli

- D. The ductus arteriosus serves to move deoxygenated blood from the right side of the heart to the systemic circulation during fetal development
 - E. Normal blood oxygen saturation immediately at birth should be 95+%
33. Which of the following is FALSE regarding common respiratory problems in neonates?
- A. Findings on chest x-ray for neonates with meconium aspiration include normal lung volume and fluid extending between the lobes of the lung
 - B. Preterm infants are at greater risk of respiratory distress syndrome due to immaturity of surfactant production
 - C. Pneumomediastinum can be identified on a chest x-ray, lateral view
 - D. Transient tachypnea of the newborn is more common in babies delivered by Caesarian section without preceding labour
34. Neonatal sepsis risk factors include all of the following except:
- A. Preterm labour
 - B. Meconium aspiration
 - C. Fetal heart rate accelerations
 - D. Chorioamnionitis
 - E. Untreated maternal GBS
35. Management of neonatal sepsis includes all of the following except:
- A. Perform blood, urine, and CSF cultures to identify the causative agent
 - B. Chest x-ray
 - C. Broad-spectrum antibiotic use, including antivirals if HSV is suspected
 - D. Provide hemodynamic and ventilation support as necessary
 - E. Begin antibiotics once culture and sensitivity results are available
36. Which of the following is NOT one of the 4 P's that influence labour and delivery?
- A. Passage
 - B. Power
 - C. Passenger
 - D. Prematurity
 - E. Psyche
37. Which of the following is NOT true regarding induction of birth?
- A. It is acceptable to induce birth when a neonate has surpassed its expected delivery date
 - B. The cervix can be ripened using medical or mechanical means as a method of induction
 - C. Labour may be induced by artificially rupturing the membranes
 - D. In cases of maternal cardiac disease it is best to delay induction
 - E. Contractions can be initiated using IV oxytocin
38. Which of the following regarding operative vaginal delivery is FALSE?

- A. Fetal malpresentation often requires rotation and is an indication for operative vaginal delivery
 - B. Operative vaginal delivery can be safely performed in cases of incomplete cervical dilation
 - C. Atypical and abnormal fetal heart tracings are the most common indication for performing an operative vaginal delivery
 - D. Methods of operative vaginal delivery include forceps use and vacuum assistance
39. Which of the following regarding postpartum hemorrhage is FALSE?
- A. One potential cause of postpartum hemorrhage is retained products
 - B. Women with pre-existing coagulopathy, like von Willebrand disease, are at a greater risk of postpartum hemorrhage
 - C. Poor lactation is a possible complication of postpartum hemorrhage
 - D. Prolonged labour is a protective factor against postpartum hemorrhage as it increases time for hemostasis
 - E. Uterine inversion is a serious condition where the uterus turns inside out and is a potential cause of postpartum hemorrhage

Answers

1. C: hCG levels increase exponentially in early pregnancy, then plateau during the late first trimester and steadily decline until they reach a steady state in the second and third trimesters.
2. A: Once an embryo is visible, the MSD should no longer be used to estimate gestational age as CRL is the most accurate method. Options c and d are distractors and are not used to estimate gestational age.
3. C: Pelvic inflammatory disease is the commonest cause of ectopic pregnancy. Endometriosis and IUD use have also been linked with ectopic pregnancy. Adenomyosis is not a known risk factor for ectopic pregnancy.
4. B is correct. Most common site for choriocarcinoma metastases is the lungs. Partial hydatidiform moles contain 2 sets of paternal chromosomes and 1 set of maternal chromosomes. After a complete hydatidiform mole, patients should abstain from conception for at least a year after serum beta hCG levels have returned to baseline.
5. B. If the patient's mass has ruptured, surgical management is indicated. If a patient were hemodynamically unstable, chestfeeding, or non-compliant to follow-up, these would also be indications for surgical management.
6. B: First line therapy for threatened spontaneous abortion is expectant management and treatment of BV if present.
7. A, C, F. The other options are possible but less common complications of D&C.
8. B. Patients with hematometra (post-abortal syndrome) commonly present with acute pain and no vaginal bleeding and may have urinary retention as blood accumulates in the uterine cavity. The uterus should be re-evacuated immediately.

9. D. Iron and folate are both needed in significant quantities by the developing fetus and are often limited in typical diets worldwide. Calcium, zinc, and vitamin C may be indicated based on maternal issues.
10. A. The pertussis (Tdap) vaccine is recommended at 27-32 weeks to provide passive immunity to the baby as maternal antibodies cross the placenta. MMR and BCG for tuberculosis are contraindicated as they are live attenuated vaccines. Hepatitis A may be indicated in select scenarios but is not recommended for all pregnancies.
11. C. Once pregnancy has been confirmed, recommended investigations include CBC, Group and Screen, testing for: Rubella, hepatitis B, HIV, VDRL, GC/Chlamydia, and a urine culture. Investigations for gestational diabetes occurs between weeks 24 and 28 of pregnancy.
12. B. Amenorrhea is common throughout all trimesters. Mood swings and breast tenderness are more commonly experienced in the first trimester. Abdominal bloating is more common in the second and third trimesters.
13. B. Indications for NIPT: age of pregnant parent >40 years, previous child with trisomy, nuchal translucency >3.5mm, positive eFTS.
14. A, B, C, and D are all risks of chorionic villus sampling (CVS) which occurs between 11-13 weeks.
15. D. Fetal anatomy scans are not offered until 18 weeks' gestation, at which point they will be offered for all pregnancies, regardless of risk and history.
16. C. Assessment of fetal "well-being" through Biophysical Profile and assessment of amniotic fluid are considered during the third trimester. Gestational sac diameter is observed during the first trimester.
17. D. The purpose of this question is to demonstrate that the greatest sensitivity to teratogens occurs until the 10th week of gestation, as this is when most organogenesis is taking place.
18. D. Zika virus is considered the newest addition to "TORCH" infections. Toxoplasmosis is a "TORCH" infection. Hyperthermia can result in spontaneous abortion, neural tube defects, cardiovascular defects, and abdominal wall defects. Acetaminophen is generally considered safe in pregnancy and not a teratogen.
19. C. Passage is one of the 4 P's that influence labour and delivery and refers to the bony pelvis of the mother. Even if the maternal pelvic architecture is not considered favourable for childbirth, this is not a contraindication to labour.
20. D. In primiparous women, engagement occurs approximately 2 weeks prior to labour. Cervical effacement describes cervical thinning and softening. Braxton-Hicks contractions do not indicate the onset of labour as they typically begin around the 28th week. The fetal heart rate is monitored during labour and should not undergo regular decelerations in normal labour.
21. A. More frequent cervical checks increase the likelihood of chorioamnionitis. The remaining answers are true. Some breech presentations can be delivered vaginally, like extended/frank breech, while others are unfavourable, like footling breech.
22. C. The placenta is delivered during the 3rd stage of labour, usually 2-30 minutes after delivery of the fetus.
23. B. Labour dystocia describes abnormally slow progress through labour. It is not a life-threatening or concerning finding after delivery, unlike the other conditions.

24. **D.** Contraception should begin/resume at week 6 postpartum or whenever sexual activity resumes, and lactational amenorrhea from breastfeeding is not fully reliable. Per rectum medications should be avoided in 3rd and 4th degree tears. Golden yellow milk is colostrum, which is produced exclusively in the first several days of breastfeeding postpartum, contains many different immune factors, and is not a sign of mastitis.
25. **E.** After delivery, Pam will be in a hypoestrogenic state, not hyperestrogenic, that will persist if she is breastfeeding.
26. **C.** Cases of severe postpartum hemorrhage should be referred to and cared for by obstetricians and would not be a responsibility of family physicians.
27. **B.** Midwife scope of practice varies across Canada and the world and depends on the midwife's location, training, and regulation.
28. **D.** Nipple stimulation increases oxytocin release. The posterior pituitary releases oxytocin to facilitate milk let-down. The anterior pituitary releases prolactin, which drives lactogenesis. Placental delivery causes a decrease in progesterone, which drives an increase in prolactin and lactogenesis.
29. **D.** Breastfeeding can prolong postpartum anovulation and provide some contraceptive benefit but should not be relied on. Contraception should be implemented 6 weeks postpartum or whenever sexual activity resumes.
30. **E.** Estrogen-containing contraceptives should be avoided in breastfeeding as it decreases the quantity and quality of milk production.
31. **D.** Ventilation is the most important focus of NRP. Of all babies born, 10% will need some resuscitation, and of those, 90% will do well with effective positive pressure ventilation.
32. **C.** Surfactant is made in the lungs to decrease surface tension within the alveoli to prevent their collapse. The foramen ovale closes after birth due to decreased pulmonary vascular resistance. The ductus arteriosus serves to move nutrient-rich oxygenated blood from the pulmonary trunk to the aorta during development, allowing the blood to bypass the non-functioning lungs. Normal blood oxygen saturation immediately at birth is 60-65% and normalizes after ~10 minutes.
33. **A.** A chest x-ray in a neonate with normal lung volume and fluid extending between the lobes of the lung describes the x-ray findings of a neonate with transient tachypnea of the newborn, not meconium aspiration.
34. **C.** Fetal decelerations are a risk factor for neonatal sepsis, not accelerations.
35. **E.** Broad-spectrum antibiotics should be initiated immediately if neonatal sepsis is suspected, and can be adjusted after culture and sensitivity results are available.
36. **D.** Prematurity is not one of the 4 P's. A premature baby is likely a smaller Passenger, which is one of the 4 P's.
37. **D.** Maternal cardiac disease is an indication for induction.
38. **B.** Incomplete cervical dilation is an absolute contraindication for operative vaginal delivery.
39. **D.** Prolonged labour can cause uterine muscle exhaustion and is a Tone cause of postpartum hemorrhage. Postpartum hemorrhage causes fall under the 4 T's: Tone, Tissue, Trauma, and Thrombin. Retained products is a Tissue cause of postpartum hemorrhage. von Willebrand disease is a Thrombin cause of postpartum hemorrhage. Poor lactation following postpartum

hemorrhage may represent Sheehan's syndrome where the pituitary gland infarcts and impacts prolactin-producing cells. Uterine inversion is a Trauma cause of postpartum hemorrhage.

Week 7

Questions

Hemorrhagic shock:

1. Which of the following is part of the lethal triad of hypovolemic hemorrhagic shock (choose 3):
 - a. Acidosis
 - b. Coagulopathy
 - c. Change in mental status
 - d. Hypothermia
 - e. Hyperlactatemia
 - f. Oliguria
2. Which of the following should you NOT do for a patient in active hemorrhagic shock?
 - a. Switch patient to a full liquid diet to replenish fluids
 - b. Order CBC, lyte, BUN, group and screen, coagulation study, Cr
 - c. Insert a foley catheter
 - d. Call for help and ask for gynecology consult

Antepartum hemorrhage:

3. Which of the following options correctly describes placental abruption, placental previa, placental accreta, or vasa previa (choose 1)?
 - a. Vasa previa involves placental invasion into the myometrium, resulting in a need for hysterectomy
 - b. Placental abruption can reliably be diagnosed using the Apt test paired with ultrasound.
 - c. Preterm prelabour rupture of membranes will result in the leaking of clear fluid and a positive Kleihaur-Betke test.
 - d. The treatment for placental accreta is almost always hysterectomy.
4. Which statement is FALSE (choose 1)?
 - a. One of the stat investigations to order if there is obstetrical bleed include liver function tests and creatinine.
 - b. Cervical dilatation should not be immediately assessed in antepartum hemorrhage
 - c. The blood resulting from placental previa is fetal in origin.
 - d. Ongoing bleeding can result in reduced clotting abilities (i.e. consumptive disseminated intravascular coagulopathy).

Preterm Birth

5. What is the purpose of tocolysis in threatened preterm labour?
 - a. Assist with fetal lung maturity
 - b. Assist with fetal neuroprotection and prevent possible eclampsia-related seizures (e.g. decreasing risk of cerebral palsy)

- c. Slowing down active contractions
 - d. Reduce the risk of gram-positive sepsis in neonates
6. How can preterm birth be prevented?
- a. Vaginal progesterone
 - b. Cervical cerclage
 - c. Treating infectious causes of cervicitis
 - d. All of the above

Hypertension in Pregnancy

7. A 40 y.o pregnant mom (in third trimester) presents with headache, blurry vision, fatigue, malaise, nausea/vomiting. She is found to have high liver enzymes, low plt, BP: 130/82. What is her most likely diagnosis, which should not be missed?

- a. HELLP
 - b. Pre-eclampsia
 - c. Theca lutein cyst
 - d. Molar pregnancy
8. Which of the following is FALSE (choose 1)?
- a. Comorbid conditions (e.g. diabetes, kidney disease) may increase the risk of severity or progression of pre-eclampsia.
 - b. Gestational hypertension can be diagnosed as early as 18 weeks.
 - c. Diuretics and ACEi/ARB are contraindicated in hypertensive disorder of pregnancy (HDP).
 - d. Severe HDP can result in all of the following: visual disturbances, strokes, cortical blindness, IUGR, oligohydramnios, and hepatic dysfunction.

IUGR:

9. Which of the following is the correct order for the progression of placental microvascular damage found on umbilical artery doppler?

- a. Reversed EDF → Decreased baseline EDF → absent EDF
 - b. Decreased baseline EDF → absent EDF → reversed EDF
 - c. Reversed EDF → decreased baseline EDF → absent EDF
10. Which of the following is FALSE (choose 1)?
- a. After 20 weeks, the symphysis-fundus height (SFH) correlates to the gestational age (in weeks) so if the SFH is more than 3 cm less than the gestational age, IUGR can be diagnosed.
 - b. Very low birth weight is <1500 g and may require intensive care, while extremely low birth weight is <1000 and may require intubation or NG tubes.
 - c. Premature babies are defined as small for gestational age (<10%th ile).
 - d. If a baby is less than 32 weeks gestational age, delivery is not indicated unless abnormal NST or abnormal UA dopplers.

Answers

1. A, B, D are the lethal triad! However, all of the above are indeed signs of hypovolemic shock. Treat the lethal triad by maintaining tissue perfusion (O₂, IV fluid, transfusion), warming the patient, and limiting blood loss.
2. A. Patient in hypovolemic hemorrhagic shock should be kept NPO instead. Answers B-D should all be done. Ensure to take vitals, maintain ABC, and consult gynecology urgently if the cause is gynecological. Inserting a foley catheter would be important for monitoring urine output.
3. D is correct. A is incorrect -- it describes placental accreta instead of vasa previa (vasa previa = a condition in which fetal blood vessels cross or run near the internal opening of the uterus). B is incorrect -- placental abruption is actually a diagnosis of exclusion. Ultrasound can be used but we learned in module it isn't always reliable because the collection of blood may have already drained from the uterus. C is incorrect -- preterm prelabour rupture of membranes will indeed result in leakage of clear amniotic fluid. However, Kleihaur-Betke test shows when fetal blood entered maternal circulation! PPRM wouldn't have a positive Kleihaur-Betke test. D is correct -- placental accreta will compromise the uterine myometrium so requires hysterectomy. It doesn't "always" result in hysterectomy because sometimes invasion is local.
4. C. Blood from placental previa is actually maternal in origin. A, B and D are all true. A is true because Creatinine would show kidney function and a liver function test would rule out pre-eclampsia or acute liver injury. B is true because an ultrasound needs to be performed first to confirm the antepartum hemorrhage is not a placental previa (or else assessing for cervical dilation would exacerbate the bleed)
5. C. Tocolysis slows down active contractions to facilitate transfer to tertiary center and/or administration of antenatal corticosteroids. Answer A is the function of antenatal corticosteroids. C is the function of magnesium sulfate. D is the function of penicillin.
6. D - all of the above. Cervicitis (e.g. gonorrhea, chlamydia, bacterial vaginosis, etc.) is a risk factor for preterm birth. Cervical cerclage reduces risk of preterm birth due to cervical insufficiency.
7. A. The patient most likely has HELLP, which stands for hemolysis elevated liver enzymes low plt. Note that it is an typical form of pre-eclampsia/eclampsia so doesn't require HTN for dx. Molar pregnancy (D) would likely present with extremely high hCG and painless vaginal bleeding. Theca lutein cyst (c) may present with pelvic heaviness, maternal virilization, and ruptured cysts may cause bleeding. Both molar pregnancy and theca lutein cyst would present in earlier pregnancy rather than third trimester and both are risk factors for pre-eclampsia. The patient doesn't have clinical hypertension/pre-eclampsia though because her BP reading is not over 140/90.
8. B. Gestational hypertension is diagnosed after 20 weeks (or else it is pre-existing HTN). C is TRUE -- diuretics are less effective during pregnancy because pregnancy physiology predisposes to fluid retention. ACEi/ARB are toxic to the fetal kidney. All of the adverse effects of HDP are true in answer D.
9. B. The first sign of placental microvascular damage is a drop in baseline EDF due to increased resistance to flow. Absent EDF occurs when 50% microvascular structures are damaged. Reversed EDF occurs when 70% of microvasculature is damaged.

10. C. Small for gestational age is not defined by a specific week of gestation. IUGR, SGA and premature are not synonymous terms. A, B and D are correct. D is correct because <32 wks, babies should be monitored in an attempt to remain in utero.